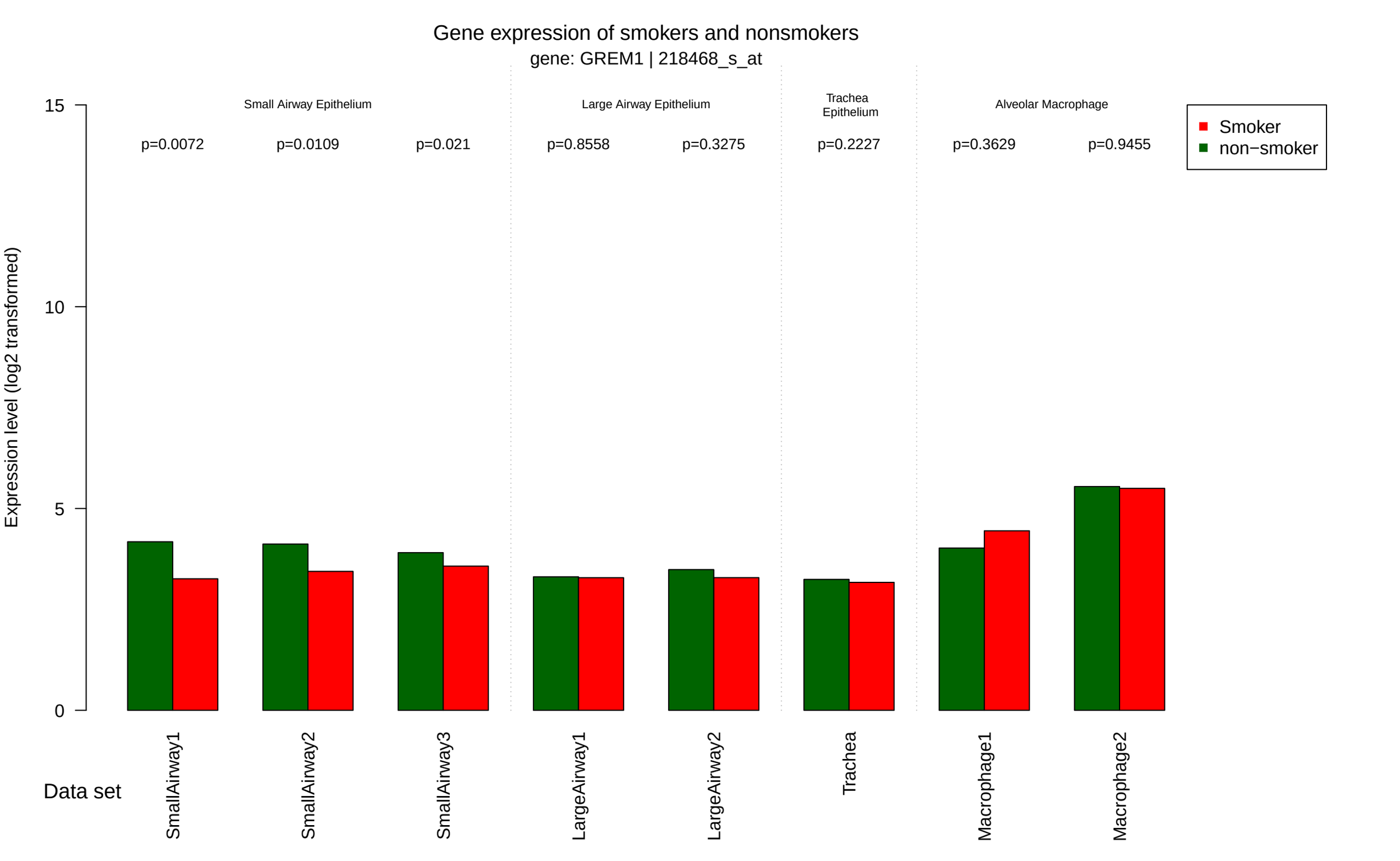
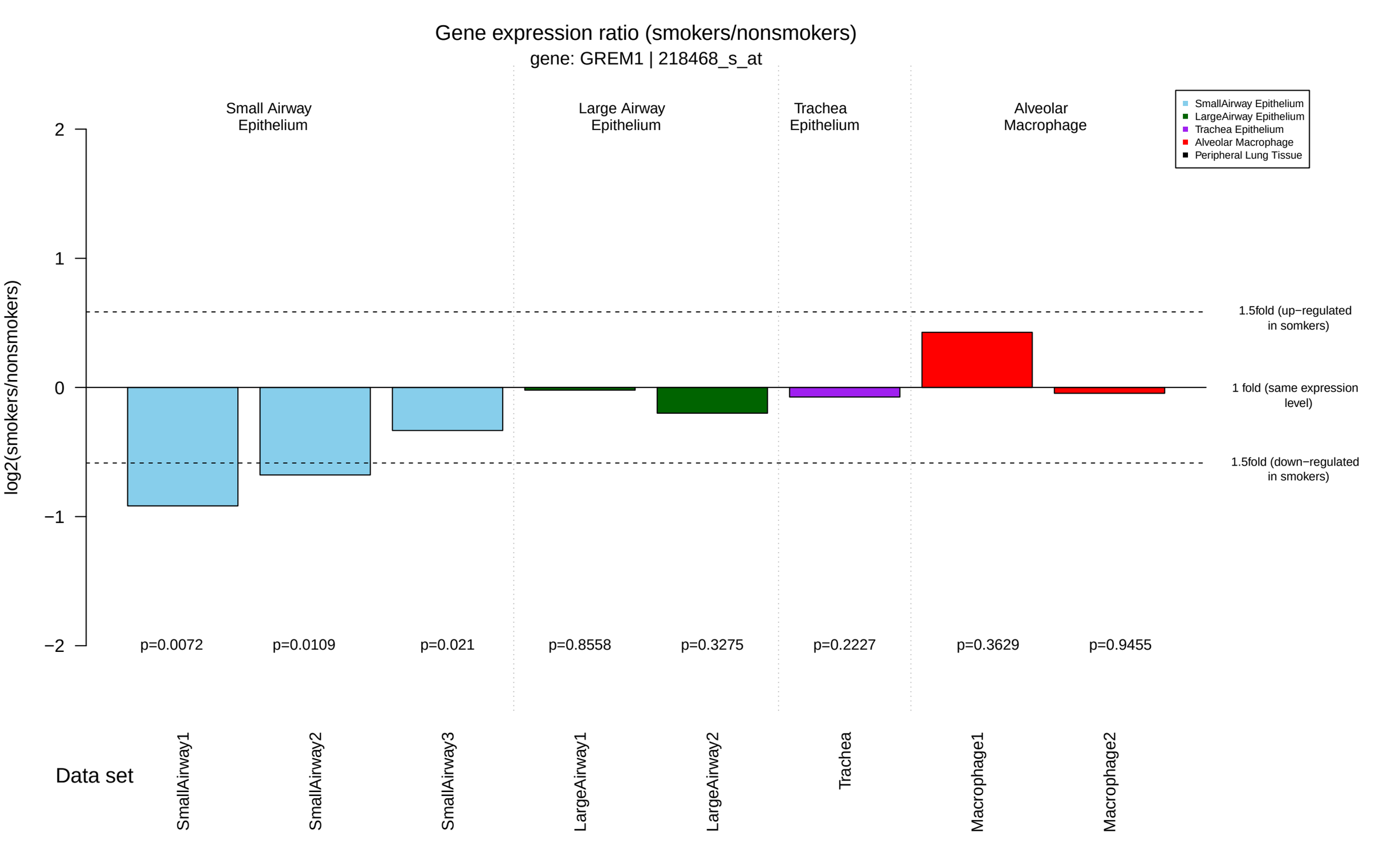
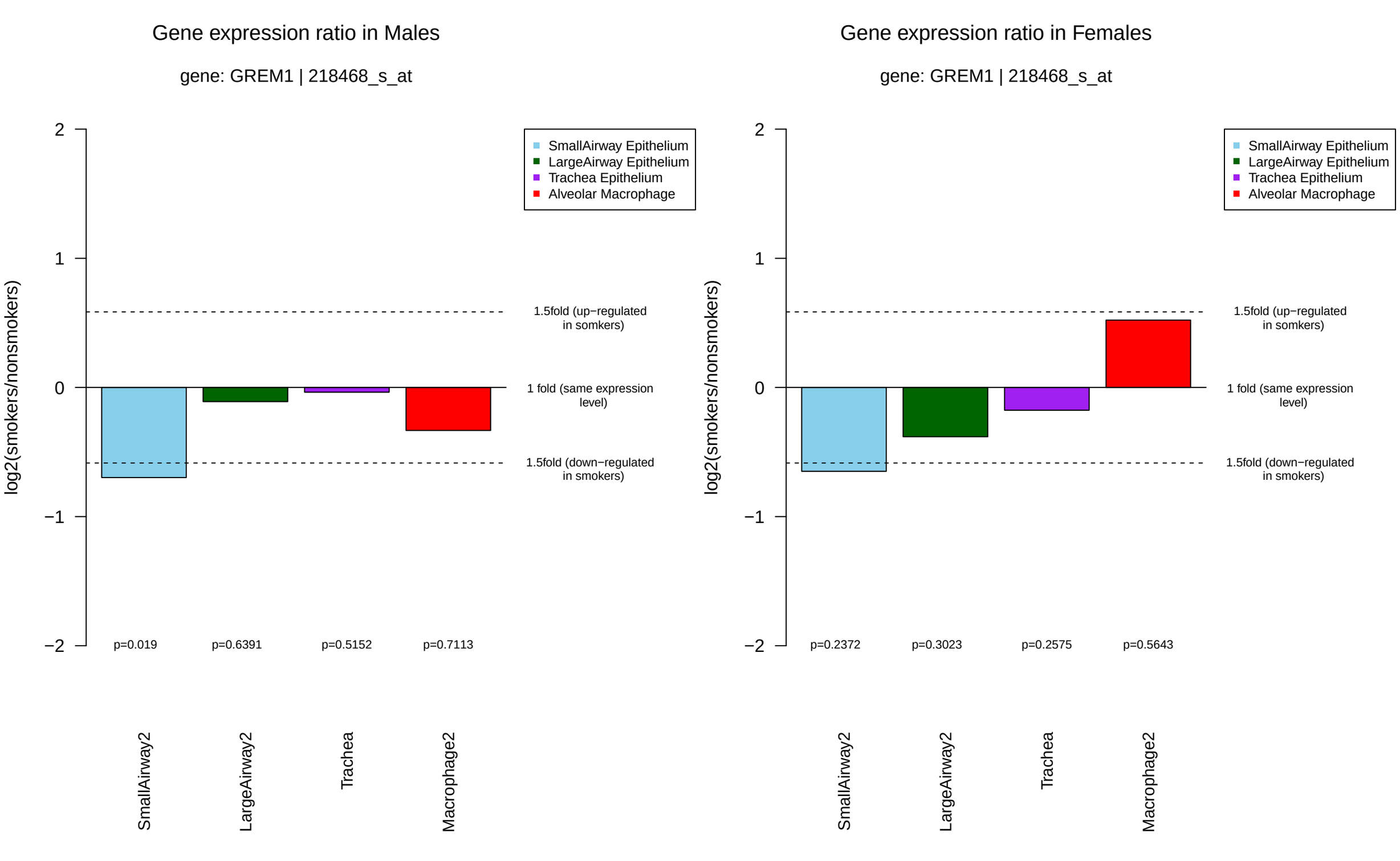
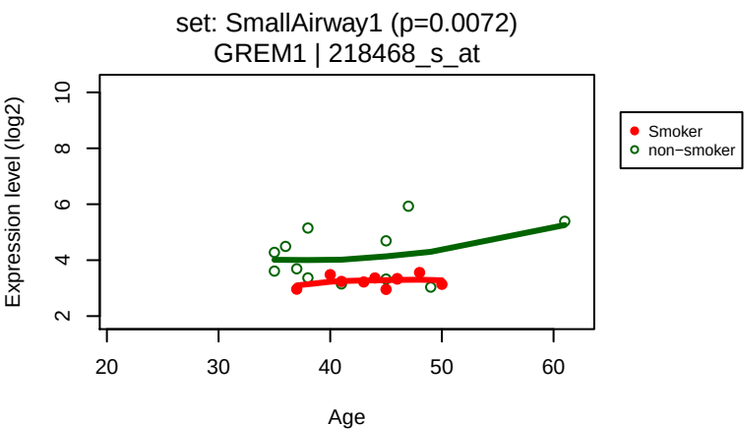
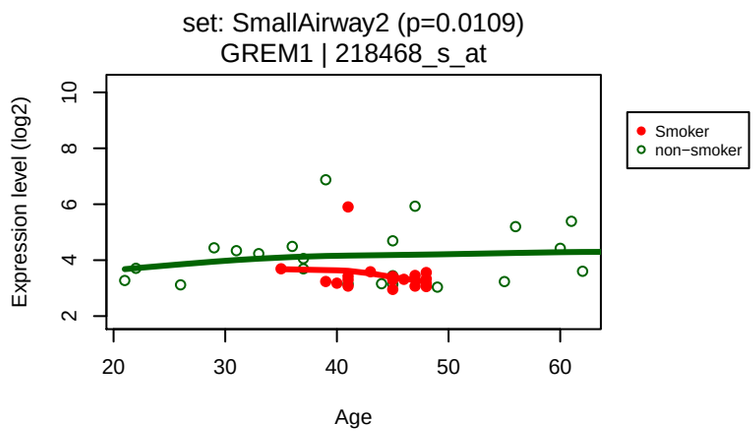
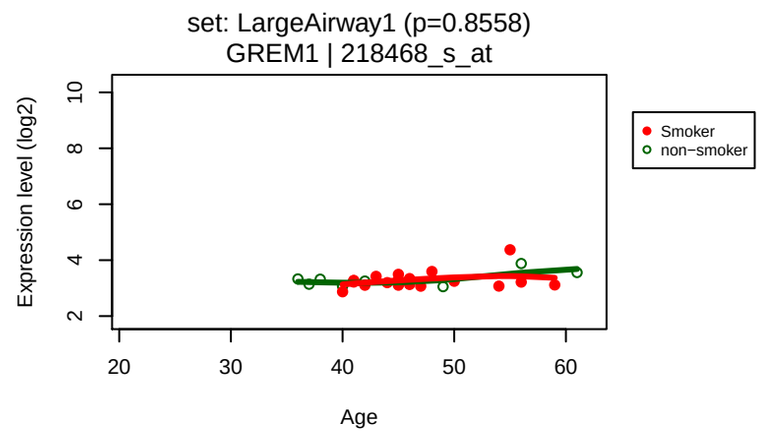

Gene expression of smokers and nonsmokers
gene: GREM1 | 218468_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0072
p=0.0109
p=0.021
p=0.8558
p=0.3275
p=0.2227
p=0.3629
p=0.9455
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

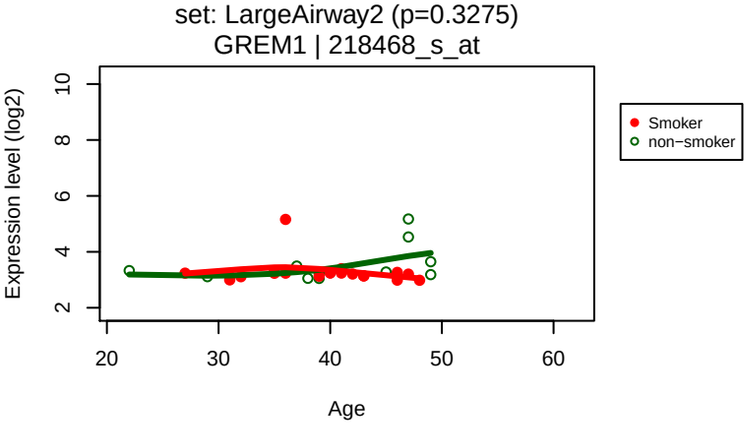

Gene expression ratio (smokers/nonsmokers)
gene: GREM1 | 218468_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0072
p=0.0109
p=0.021
p=0.8558
p=0.3275
p=0.2227
p=0.3629
p=0.9455
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

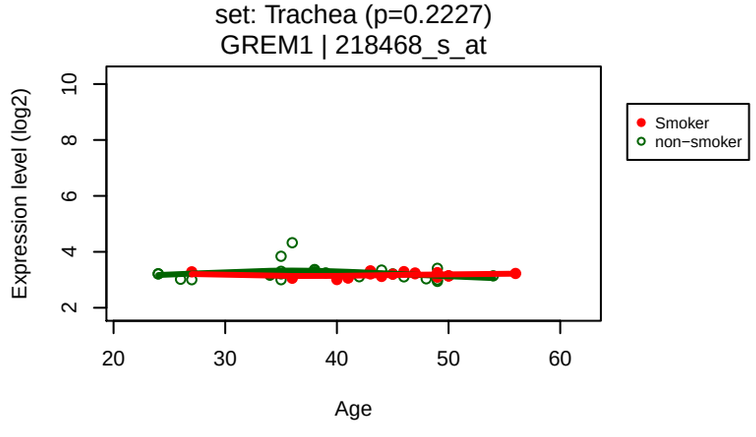

Gene expression ratio in Males
gene: GREM1 | 218468_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.019
p=0.6391
p=0.5152
p=0.7113
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GREM1 | 218468_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2372
p=0.3023
p=0.2575
p=0.5643
SmallAirway2
LargeAirway2
Trachea
Macrophage2

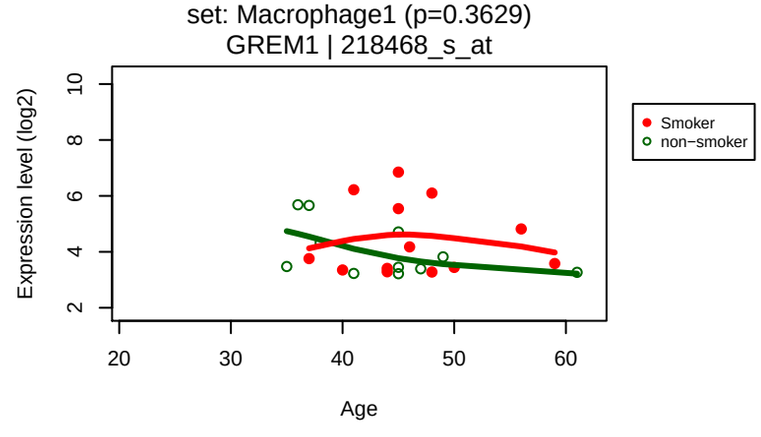

set: SmallAirway1 (p=0.0072)
GREM1 | 218468_s_at
Expression level (log2)
Age
Smoker
non−smoker

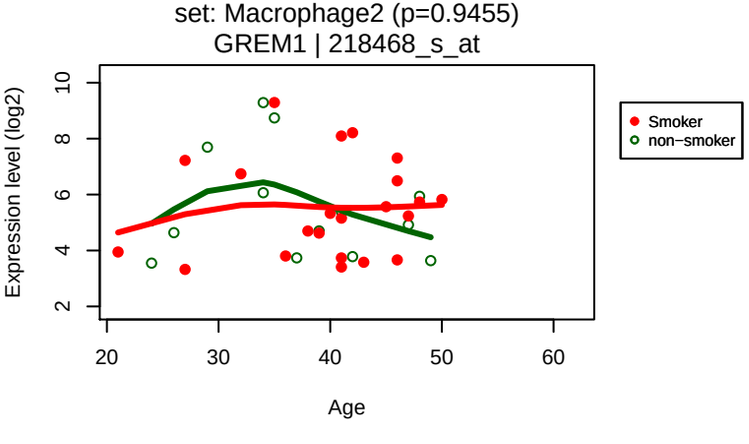

set: SmallAirway2 (p=0.0109)
GREM1 | 218468_s_at
Expression level (log2)
Age
Smoker
non−smoker

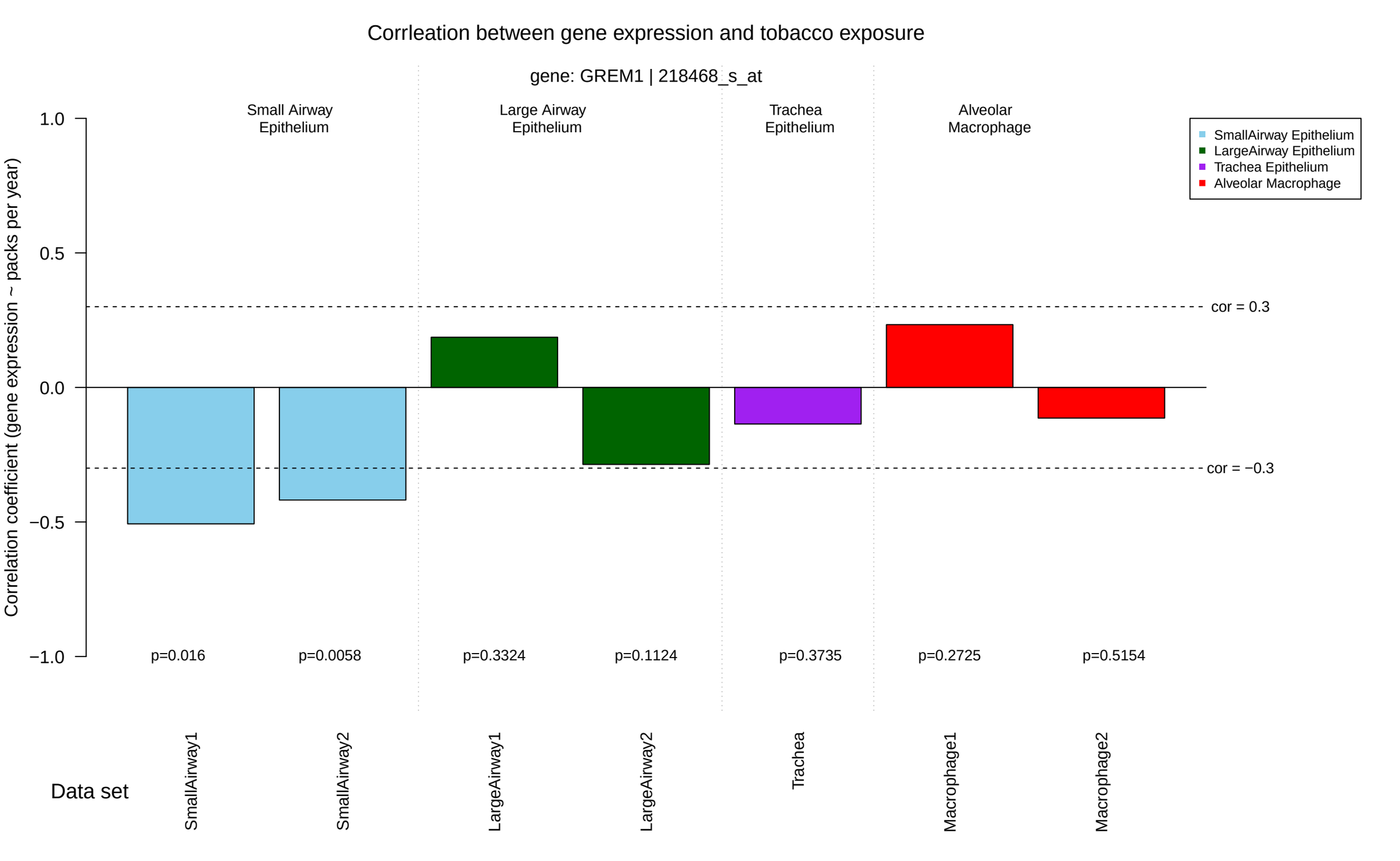

set: LargeAirway1 (p=0.8558)
GREM1 | 218468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3275)
GREM1 | 218468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2227)
GREM1 | 218468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3629)
GREM1 | 218468_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9455)
GREM1 | 218468_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GREM1 | 218468_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.016
p=0.0058
p=0.3324
p=0.1124
p=0.3735
p=0.2725
p=0.5154
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2